

Research Summary — EPFL Master's Application

Program: Master of Science in Quantum Science and Engineering

Applicant: Ananya Asthana

Overview

This research portfolio examines how the principles of quantum mechanics can enhance computation, communication, and cryptographic security. It focuses on the design of hybrid quantum–classical frameworks that merge the reliability of established cryptography with the scalability of quantum systems. The projects combine simulation, algorithmic modeling, and analytical exploration, emphasizing the translation of theoretical constructs into deployable technologies.

Hybrid Quantum–Classical System Development

The project Hybrid QKD + AES-256 Secure Communication System integrates Quantum Key Distribution (QKD) with AES-256 encryption to form an adaptive, hybrid communication layer. Built using SimulaQron and NS-3, the system monitors and regulates the Quantum Bit Error Rate (QBER) in real time to ensure key validity prior to encryption.

Simulations demonstrated stable key synchronization under varying link conditions — [insert QBER and key-rate metrics here] — and validated sifting, reconciliation, and privacy-amplification stages. The model provides a measurable foundation for assessing the resilience of hybrid cryptographic infrastructures and their performance across noisy channels.

Computational Simulations and Algorithmic Modeling

Complementary Qiskit-based simulations investigate quantum-algorithm design and execution fidelity. The portfolio includes implementations of entanglement protocols, quantum teleportation, Grover's search, and the Quantum Fourier Transform (QFT). These circuits are benchmarked for logical coherence and gate-level performance — [insert fidelity or error metrics here] — supporting studies of qubit interaction, measurement optimization, and gate decomposition.

Together, these experiments function as a diagnostic suite for analyzing algorithmic scalability and hardware-agnostic error behavior in mid-scale quantum processors.

Independent Research and Theoretical Analyses

Independent analytical work explores emerging paradigms in quantum computation and information theory, including Quantum Neural Networks (QNNs) for optimization, neglecton-based braiding for topological fault tolerance, and quantum–AI synergy in adaptive control. These studies assess how novel quantum constructs may enhance computational reliability and inform secure, data-driven system design. [insert model comparison or resource-scaling notes here]

Future Research Directions

Current efforts aim to develop secure and scalable quantum-communication networks that employ quantum correlations (including, but not limited to, entanglement) alongside post-quantum cryptography (PQC) and classical encryption. The goal is to achieve measurable improvements in QBER reduction ($\geq 50\%$) and key-refresh rates (~ 10 kHz) under realistic network conditions.

At EPFL’s Laboratory for Security and Cryptography (LASEC), this work will progress toward modular prototypes that dynamically allocate quantum hardware to high-security layers while deploying PQC across broader infrastructures — enabling communication systems that are simultaneously quantum-capable and quantum-resilient.

Outlook

Subsequent phases will extend this framework toward quantum-algorithm optimization and hybrid computational models, linking secure communication with high-performance computation. The overarching objective is to establish quantum-resilient infrastructures that support both data integrity and scientific discovery in the emerging quantum-internet era.

Alignment with EPFL

EPFL’s interdisciplinary ecosystem—where cryptography, computation, and experimental quantum systems intersect—provides the ideal environment to advance this line of inquiry. The Master’s in Quantum Science and Engineering offers the academic depth and collaborative platform required to transform these frameworks into practical, scalable implementations aligned with EPFL’s mission to pioneer secure quantum technologies.